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Semiconductor memory device

Abstract

A semiconductor memory device includes a substrate, bit lines, contacts, a dielectric layer, storage node pads and a capacitor structure. The bit lines are disposed on the substrate and include a plurality of first bit lines and at least one second bit line. The contacts are disposed on the substrate and alternately and separately disposed with the bit lines. The dielectric layer is disposed over the contacts and bit lines. The storage node pads are disposed in the dielectric layer and respectively contact the contacts. The capacitor structure is disposed on the storage node pads and includes a plurality of first capacitors and at least one second capacitor located above at least one second bit line. Therefore, the semiconductor memory device can achieve more optimized device performance.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6194758	12/2000	Tanaka	N/A	N/A
6313497	12/2000	Iwasa	N/A	N/A
8023352	12/2010	Masuoka	365/210.15	H10B 12/482
9184168	12/2014	Ryu	N/A	N/A
9960167	12/2017	Ho	N/A	N/A
2020/0365537	12/2019	Choi	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111968977	12/2019	CN	H01L 27/10823

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/688,858, filed on Mar. 7, 2022. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a semiconductor memory device and the method for forming thereof, in particular to a semiconductor memory device including dynamic random access memory and the method for forming the same.

2. Description of the Prior Art

(2) With the trend of miniaturization of various electronic products, the design of semiconductor memory devices must meet the requirements of high integration and high density. For a dynamic

random access memory (DRAM) having recessed gate structures, because the carrier channel of which is relatively long in the same semiconductor substrate compared with that of the DRAM without recessed gate structures, the leakage current from the capacitor structure in the DRAM can be reduced. Therefore, the DRAM having recessed gate structures has gradually replaced DRAM having planar gate structures under the current mainstream development trend.

(3) Generally, the DRAM having recessed gate structure is constructed by a large number of memory cells which are arranged to form an array area, and each of the memory cells can be used to store information. Each memory cell may include a transistor element and a capacitor element connected in series, which is configured to receive voltage information from word lines (WL) and bit lines (BL). In order to fulfill the requirements of advanced products, the density of memory cells in the array area must be further increased, which increases the difficulty and complexity of related fabricating processes and designs. Therefore, the present technology needs further improvement to effectively improve the performance and reliability of related memory devices.

SUMMARY OF THE INVENTION

(4) An object of the present invention is to provide a method for forming a semiconductor memory device, in which a plurality of storage node pads are formed on a substrate through a self-aligned double patterning process or a self-aligned reverse patterning process. In this way, a subsequently formed capacitor structure may include at least a portion of capacitor located on a dummy bit line to become a dummy capacitor. With this arrangement, the semiconductor memory device may be formed on the prerequisite of simplifying the manufacturing process, and is isolated from the surrounding active devices through the arrangement of the dummy capacitor, thereby achieving the effect of optimizing the overall device performance.

(5) To achieve the above objective, one embodiment of the present invention provides a semiconductor memory device, which includes a substrate, a plurality of bit lines, a plurality of contacts, a dielectric layer, a plurality of storage node pads and a capacitor structure. The bit lines are disposed on the substrate and include a plurality of first bit lines and at least one second bit line, and the at least one second bit line is disposed outside all the first bit lines. The contacts are disposed on the substrate and alternately and separately disposed with the bit lines. The dielectric layer is disposed over the contacts and the bit lines. The storage node pads are disposed in the dielectric layer and respectively contact the contacts. The capacitor structure is disposed on the storage node pads, and includes a plurality of first capacitors which are respectively aligned to the storage node pads, and at least one second capacitor which is located above the at least one second bit line.

(6) To achieve the above object, one embodiment of the present invention provides a method for forming a semiconductor memory device, which includes the following steps. First, a substrate is provided, and a plurality of bit lines are formed on the substrate, in which the bit lines include a plurality of first bit lines and at least one second bit line, and the at least one second bit line is formed outside all the first bit lines. Then, a plurality of contacts are formed on the substrate, the bit lines and the contacts are alternately arranged with each other, and a dielectric layer is formed above the contacts and the bit lines to cover the contacts and the bit lines. Then, a plurality of storage node pads are formed in the dielectric layer, and the storage node pads are respectively aligned to the contacts. Then, a capacitor structure is formed on the storage node pads, in which the capacitor structure includes a plurality of first capacitors respectively aligned to the storage node pads, and at least one second capacitor located above the at least one second bit line.

(7) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. **1-9** are schematic diagrams showing steps of a method for forming a semiconductor memory device in one preferred embodiment of the present invention, in which:
- (2) FIG. **1** is a schematic top view of a semiconductor memory device after forming bit lines.
- (3) FIG. **2** is a schematic cross-sectional view along the cut line A-A' in FIG. **1**;
- (4) FIG. **3** is a schematic top view of a semiconductor memory device after performing a self-aligned double patterning process;
- (5) FIG. **4** is a schematic cross-sectional view along the cut line A-A' in FIG. **3**;
- (6) FIG. **5** is a schematic top view of a semiconductor memory device after forming storage node pads;
- (7) FIG. **6** is a schematic cross-sectional view along the cut line A-A' in FIG. **5**;
- (8) FIG. **7** is a schematic cross-sectional view along the cut line B-B' in FIG. **5**;
- (9) FIG. **8** is a schematic cross-sectional view of a semiconductor memory device after forming a stacked structure;
- (10) FIG. **9** is a schematic cross-sectional view of a semiconductor memory device after forming a bottom electrode layer;
- (11) FIG. **10** is a schematic cross-sectional view of a semiconductor memory device after forming a top electrode layer; and
- (12) FIG. **11** is another schematic cross-sectional view of a semiconductor memory device after forming the top electrode layer.
- (13) FIG. **12** is a schematic cross-sectional view of a semiconductor memory device according to a second embodiment of the present invention.
- (14) FIG. **13** is a schematic sectional view of a semiconductor memory device according to a third embodiment of the present invention.
- (15) FIG. **14** is a schematic sectional view of a semiconductor memory device according to a fourth embodiment of the present invention.

DETAILED DESCRIPTION

- (16) For better understanding of the present invention, some embodiments of the present invention are listed below with the accompanying drawings, the composition and the desired effects of the present invention are described in detail for those skilled in the art. In addition, the technical features in different embodiments described in the following may be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.
- (17) Please refer to FIGS. **1-8**, which are schematic diagram showing steps of a method for forming a semiconductor memory device **300** in a first embodiment of the present invention. First, as shown in FIG. **1**, a substrate **100** is provided, such as a silicon substrate, a silicon-containing substrate (such as SiC, SiGe) or a silicon-on-insulator (SOI) substrate, and at least one insulation area **101**, such as a shallow trench isolation (STI), is formed in the substrate **100** to define a plurality of active areas (AAs) **103** on the substrate **100**. Preferably, a plurality of active areas **103** extend parallel to each other and spaced apart from each other along a first direction **D1**, and are alternately disposed with each other, in which the first direction **D1** intersects and is not perpendicular to the y direction or the x direction, as shown in FIG. **1**. In one embodiment, the insulation area **101** is formed by etching a plurality of trenches (not shown in the drawing) in the substrate **100**, and then filling the trenches with an insulating material (such as silicon oxide or silicon oxynitride), but not limited thereto.
- (18) In addition, a plurality of buried gates (not shown in the drawing) may be formed in the substrate **100**, for example, the buried gates extend parallel to each other along a direction (such as y direction) and cross the active area **103**, so as to serve as buried word lines (BWLs, not shown in the drawing) of the semiconductor memory device **300**. A plurality of bit lines **160** and a plurality

of contacts **180** (not shown in FIG. 1) may be formed on the substrate **100**. For example, the bit lines **160** respectively extend in another direction (e.g., the x direction) perpendicular to the direction and cross the active area **103**. Although the buried gates are not specifically depicted in the drawings of the present embodiment, it can be easily understood by those skilled in the art that the bit lines **160** extending in the x direction should be perpendicular to the buried gates extending in the y direction from a top view, and the bit lines **160** cross the active area **103** and the buried gates at the same time.

(19) As shown in FIG. 1, the bit lines **160** further includes a plurality of first bit lines **162** and at least one second bit line **164**. The first bit lines **162** and the at least one second bit line **164** are respectively disposed in a memory cell region **300a** and a periphery region **300b** of the semiconductor memory device **300**, and may be used as general bit lines (BLs) and dummy bit lines (dummy BLs) respectively. In particular, the at least one second bit line **164** may be located at a side outside all the first bit lines **162**, but not limited thereto. It can be easily understood by those skilled in the art that, under actual requirements of devices, the memory cell region **300a** and the periphery region **300b** may also have other types of arrangement, and thus the first bit lines and the at least one second bit line may be adjusted to have other arrangement patterns or the at least one second bit line may be adjusted to have other numbers. For example, in one embodiment, the semiconductor memory device preferably includes two second bit lines **164**, which are respectively arranged at two opposite sides (i.e., upper and lower sides) of all the first bit lines **162** to isolate other external devices. Furthermore, in the present embodiment, a line width W2 of each second bit line **164** (for example, the width in the y direction) is preferably larger than a line width W1 of each first bit line **162**, but it is not limited thereto. In another embodiment, the at least one second bit line and the first bit lines may also be selectively made to have the same line width.

(20) In detail, as shown in FIG. 2, each bit line **160** is formed on the substrate **100** apart from each other and includes a semiconductor layer (e.g., polysilicon) **161**, a barrier layer **163** (e.g., titanium and/or titanium nitride), a conductive layer **165** (e.g., metal with low resistance such as tungsten, aluminum or copper), and a cap layer **167** (e.g., silicon oxide, silicon nitride or silicon oxynitride, etc.) stacked in sequence. It should be noted that a portion of the first bit lines **162** are formed on a dielectric layer **130** above the substrate **100**, in which the dielectric layer **130** preferably has a composite layer structure, such as an oxide-nitride-oxide (ONO) structure, but not limited thereto. Another portion of the first bit lines **162** further form bit line contacts (BLCs) **160a** below each of them, which extend into the substrate **100** and directly contact the substrate **100** below (the active areas **103**). Furthermore, the bit line contacts **160a** are integrally formed with the semiconductor layer **161** of the another portion of the first bit lines **162**, as shown in FIG. 2. On the other hand, the contacts **180** are also formed on the substrate **100** separately from each other and alternately disposed with the bit lines **160**. In addition, each of the contacts **180** and each of the bit lines **160** are insulated from each other through a spacer structure **170**. In one embodiment, the spacer structure **170** may optionally have a single layer structure or a composite layer structure as shown in FIG. 2, which includes, for example, a first spacer **171** (for example, containing silicon nitride), a second spacer **173** (for example, containing silicon oxide) and a third spacer **173** (for example, containing silicon nitride) stacked in sequence, but not limited thereto. In addition, each of the contacts **180** further includes a plurality of first contacts **182** and at least one second contact **184**, wherein the first contacts **182** are disposed in the memory cell region **300a** of the semiconductor memory device **300** and directly contacts the underlying substrate **100** (including the active areas **103** and the insulation areas **101**) to serve as a storage node contact (SNC) of the semiconductor memory device **300**, and the second contact **184** is disposed in the periphery region **300b** of the semiconductor memory device **300**. In one embodiment, the contacts **180** include low resistance metal materials such as aluminum (Al), titanium (Ti), copper (Cu) or tungsten (W).

(21) Then, a plurality of storage node pads (SN pads) **211** are formed on the substrate **100**. In particular, the storage node pads **211** are formed by a self-aligned double patterning (SADP)

process or a self-aligned reverse patterning (SARP) process, but not limited thereto. Referring to FIG. 3 and FIG. 4, a metal layer **210** is formed above the contacts **180** and the bit lines **160**. The metal layer **210** is made of a low-resistance metal material such as aluminum, titanium, copper or tungsten, preferably a metal material different from the contacts **180**, but not limited thereto. Then, at least two self-aligned double patterning processes are sequentially performed to form a plurality of patterned masks **221** extending parallel to each other along a second direction D2 and a plurality of patterned masks **223** extending parallel to each other along a second direction D3 on the metal layer **210**. In particular, the second direction D2 and the second direction D3 intersect with each other and are not perpendicular to the y direction or the x direction respectively, for example, an included angle between the second direction D2 or the second direction D3 and the y direction or the x direction is about 60 to 120 degrees, as shown in FIG. 3, but not limited thereto. Then, the relative positions of the storage node pads **211** are defined by the overlapped portions between the patterned masks **221** and the patterned masks **223**, and patterned masks **221**, **223** are formed as etching masks to perform an etching process to pattern the underlying metal layer **210**, and thus the storage node pads **211** may be formed, as shown in FIGS. 5-6. It should be noted that, the storage node pads **211** are formed by controlling the overlapping portions of the patterned masks **221** and the patterned masks **223**, and a portion of the patterned masks **223** are not extended over the memory cell region **300a** to form a serrated edge. Then, the storage node pads **211** formed accordingly in the present embodiment are selectively above and aligned to each of the first contacts **182** within the memory cell region **300a** as shown in FIGS. 5-6, and not above the second contact **184** within the periphery region **300b** as shown in FIGS. 5-6, and not above the at least one second bit line **164** as shown in FIGS. 5, 7.

(22) Subsequently, a capacitor structure **250** may be formed above the storage node pads **211** to directly contact and electrically connect with the storage node pads **211** below. In one embodiment, the manufacturing process of the capacitor structure **250** includes but not limited to the following steps. First, as shown in FIG. 8, a dielectric layer **230** and a supporting layer structure **240** are formed on the substrate **100**. The dielectric layer **230** is disposed over the contacts **180** (including the first contacts **182** and the second contact **184**) and the bit lines **160**. Preferably, the thickness of the dielectric layer **230** is greater than the thickness of each of the storage node pads **211**, and thus the storage node pads **211** may be located in the dielectric layer **230**. The supporting layer structure **240** includes, for example, at least one oxide layer and at least one nitride layer alternately stacked. In the present embodiment, the supporting layer structure **240** includes, for example, a first supporting layer **241** (for example, including silicon oxide), a second supporting layer **243** (for example, including silicon nitride or silicon carbonitride), a third supporting layer **245** (for example, including silicon oxide) and a fourth supporting layer **247** (for example, including silicon nitride or silicon carbonitride, etc.) stacked in sequence from bottom to top, but not limited thereto. Preferably, the first supporting layer **241** and the third supporting layer **245** may have relatively large thicknesses, for example, about 5 times to 10 times more than other supporting layers (the second supporting layer **243** or the fourth supporting layer **247**), but not limited thereto. Therefore, the overall thickness of the supporting layer structure **240** may reach but not limited to about 1600 angstroms to about 2000 angstroms. It should be understood by those skilled in the art that the specific stacking number of the aforementioned oxide layer (such as the first supporting layer **241** or the third supporting layer **245**) and the aforementioned nitride layer (such as the second supporting layer **243** or the fourth supporting layer **247**) is not limited to the aforementioned number, but may be adjusted according to actual requirements, such as 3 layers, 4 layers or other numbers. Then, a plurality of first openings **242**, at least one second opening **246** and at least one third opening **244** are formed in the supporting layer structure **240**, all of which sequentially penetrate through the fourth supporting layer **247**, the third supporting layer **245**, the second supporting layer **243**, the first supporting layer **241** and part of the dielectric layer **230**. The first openings **242** are disposed within the memory cell region **300**, to respectively align to the storage

node pads **211** (and the first contacts **182**) underneath, so that the top surfaces of the storage node pads **211** may be exposed from each of the first openings **242**. The at least one third opening **244** is disposed within the periphery region **300b** to align to the second contact **184** underneath. However, because the bottom surfaces of the third opening **244** is lower than the top surface of the dielectric layer **230** without penetrating the dielectric layer **230**, only part of the dielectric layer **230** is exposed from the at least one third opening **244**. The at least one second opening **246** is also disposed within the periphery region **300b**, outside all the first openings **242** and the third opening **244**, wherein the at least one second opening **246** is aligned to the at least one second bit line **164** located in the periphery region **300b**. The at least one second opening **246** also does not penetrate through the dielectric layer **230** to partially expose the dielectric layer **230** therefrom, as shown in FIG. 8.

(23) Then, as shown in FIG. 9, bottom electrode layers **251** are formed to fill the first openings **242**, the at least one third opening **244** and the at least one second opening **246**, respectively. In particular, each of the bottom electrode layers **251** includes a low-resistance metal material such as aluminum, titanium, copper or tungsten, and preferably titanium, but not limited thereto. It should be noted that the bottom electrode layers **251** disposed in the first openings **242** directly contact the storage node pads **211** below. The bottom electrode layers **251** disposed in the at least one third opening **244** and the at least one second opening **246** directly contact the dielectric layer **230**, and are located directly above the second contact **184** and the at least one second bit line **164**, respectively. As shown in FIG. 9, after the bottom electrode layers **251** are formed, an etching process is performed through a mask layer (not shown) to completely remove the oxide layer (such as the first supporting layer **241** or the third supporting layer **245**) in the supporting layer structure **240** and partially remove the nitride layer (such as the second supporting layer **243** or the fourth supporting layer **247**) in the supporting layer structure **240**.

(24) Subsequently, as shown in FIG. 10, a capacitor dielectric layer **253** and a top electrode layer **255** are sequentially formed on the bottom electrode layers **251**. In particular, part of the capacitor dielectric layer **253** and part of the top electrode layer **255** may be further filled between the remaining second supporting layer **243** and the fourth supporting layer **247**, and between the remaining second supporting layer **243** and the dielectric layer **230**. In one embodiment, the capacitor dielectric layer **253** includes a high dielectric constant dielectric material selected from the group consisting of hafnium oxide (HfO_2), hafnium silicon oxide (HfSiO_4), hafnium silicon oxynitride (HfSiON), zinc oxide (ZnO), titanium oxide (TiO_2) and zirconia-alumina-zirconia (ZAZ), and preferably includes zirconia-alumina-zirconia. The top electrode layer **255** includes low resistance metal materials such as aluminum, titanium, copper or tungsten, preferably titanium, but not limited thereto.

(25) Thus, the manufacturing process of the capacitor structure **250** is completed. The capacitor structure **250** includes the bottom electrode layer **251**, the capacitor dielectric layer **253**, and the top electrode layer **255** stacked in sequence, thereby forming a plurality of vertically extending capacitors **250a**, **250b**, and **250c**. It should be noted that the capacitor structure **250** includes a plurality of the first capacitors **250a** disposed in the memory cell region **300** to align to the first contacts **182**, respectively. In particular, each of the first capacitors **250a** may be electrically connected with a transistor device (not shown) of the semiconductor memory device **300** through the storage node pad **211** and a storage node plug (i.e., the first contact **182**) below, and each of the first capacitors **250a** may thereby be used as a storage node (SN) of the semiconductor memory device **300** to maintain a good contact relationship between the capacitor structure **250** and the transistor device. On the other hand, the capacitor structure **250** further includes at least one second capacitor **250b** and at least one third capacitor **250c** disposed in the periphery region **300b**. Since there is no storage node pad **211** disposed under the second capacitor **250b** and the third capacitors **250c**, the second capacitor **250b** and the third capacitors **250c** may not be electrically connected with the storage node plug (i.e., the second contact **184**) below. The bottom surfaces of the second

capacitor **250b** and the third capacitor **250c** (i.e., the bottom surfaces of the bottom electrode layer **251** filling the third opening **244** and the second opening **246**) only contact the dielectric layer **230** to form an open circuit, which become a dummy storage node (dummy SN) and are isolated from the adjacent storage nodes to maintain the overall device performance. In particular, the bottom surfaces of the second capacitor **250b** and the third capacitor **250c** are lower than the top surface of the dielectric layer **230**, as shown in FIG. **10**. In addition, the second capacitor **250b** is located outside all the first capacitors **250a** and the third capacitor **250c** to align to the second bit line **164** in the periphery region **300b**.

(26) Although the schematic cross-sectional view shown in FIG. **10** only shows that a single second capacitor **250b** is located on the second bit line **164**, and a single third capacitor **250c** is located outside the first capacitor **250a** within the periphery region **300b**, it should be easily understood by those skilled in the art that a plurality of third capacitors **250c** and a plurality of second capacitors **250b** are disposed within the periphery region **300b**. For example, if being viewed from a schematic cross-sectional view along the cross line B-B', a plurality of the second capacitors **250b** is located on the second bit line **164**, as shown in FIG. **11**. Therefore, the semiconductor memory device **300** of the present embodiment may form a dynamic random access memory (DRAM) device, in which at least one transistor device and at least one first capacitor **250a** constitute the smallest memory cell in the DRAM array to receive voltage information from the bit lines **160** and the buried word lines.

(27) Thus, the semiconductor memory device **300** in the first embodiment of the present invention is completed. According to the forming method of the present embodiment, the storage node pads **211** are formed by controlling the overlapping parts of the patterned mask **221** and the patterned mask **223**, and the storage node pads **211** are thereby disposed above the first contacts **182** within the memory cell region **300a**, but not above the second contact **184** within the periphery region **300b**. In this way, after the capacitor structure **250** is formed, the first capacitors **250a** which may be used as storage nodes and the at least one second capacitor **250b** and/or the at least one third capacitors **250c** which may be used as a dummy storage node may be respectively formed. In particular, the storage nodes (i.e., the first capacitors **250a**) are electrically connected with the transistor device (not shown) of the semiconductor storage device **300** through the storage node pads **211** and the storage node plugs (i.e., the first contact **182**) below, while the dummy storage nodes (i.e., the at least one second capacitor **250b** and/or the at least one third capacitors **250c**) are not provided with the storage node pad **211** below, and thus it may not be electrically connected with the storage node plugs (i.e., the second contact **184**) below. The arrangement of the dummy storage nodes may stabilize and improve the performance of the storage nodes, and may be used to isolate the adjacent storage nodes to maintain the overall device performance of the semiconductor storage device **300**.

(28) In addition, it should be easily understood by those skilled in the art that, on the prerequisite for meeting the actual product requirements, there may be other types for forming the semiconductor memory device and the method for forming thereof according to the present invention, which are not limited to the foregoing. For example, the dummy storage nodes may have other types optionally. Other embodiments or variations of the forming method of the semiconductor memory device in the present invention will be further described below. To simplify the description, the follow description mainly focuses on the difference of each embodiment, and will not repeat the same description. In addition, the same components in each embodiment of the present invention are labeled with the same reference numerals, so as to facilitate cross-reference among the embodiments.

(29) Please refer to FIG. **12**, which shows steps of a method for forming a semiconductor memory device **400** in a second embodiment of the present invention. The forming steps of the front end of the semiconductor memory device **400** in the present embodiment are substantially the same as those steps of the front end of the semiconductor memory device **300** in the first embodiment

described above, and will not be repeated here. The main difference between the present embodiment and the first embodiment is that at least one second capacitor **450b** penetrates through the dielectric layer **230** and may directly contact the cap layer **167** of the second bit line **164**.

(30) In detail, the forming method of the present embodiment further controls the etching process conditions when forming the openings in the supporting layer structure **240**, and selectively makes the second opening (not shown) aligned to the second bit line **164** penetrate the dielectric layer **230** and stop on the top surface of the cap layer **167** of the second bit line **164**, therefore, the cap layer **167** of the second bit line **164** may be exposed from the second opening. Subsequently, the bottom electrode layer **251**, the capacitor dielectric layer **253** and the top electrode layer **255** are sequentially formed to form the capacitor structure **450** as shown in FIG. **12**. In particular, the at least one second capacitor **450b** is aligned to the second bit line **164** in the periphery region **300b**, and extends into the dielectric layer **230** and directly contacts the top surface of the cap layer **167** of the second bit line **164** through the bottom electrode layer **251**, therefore, the at least one second capacitor **450b** only contacts the dielectric layer **230** and the cap layer **167** to form an open circuit, thereby becoming the dummy storage node. Under this arrangement, the semiconductor memory device **400** of the present embodiment may also form a dynamic random access memory device, and the adjacent storage nodes are isolated by the at least one second capacitor **450b** and the at least one third capacitor **250c**, so as to maintain its overall device performance.

(31) Please refer to FIG. **13**, which shows steps of a method for forming a semiconductor memory device **500** in a third embodiment of the present invention. The forming steps of the front end of the semiconductor memory device **500** in the present embodiment are substantially the same as the forming steps of the front end of the semiconductor memory device **300** in the first embodiment described above, and will not be repeated here. The main difference between the present embodiment and the first embodiment is that at least one second capacitor **550b** penetrates through the dielectric layer **230** and further extends into a portion of the cap layer **167** of the second bit line **164**.

(32) In detail, the forming method of the present embodiment further controls the etching process conditions when forming the openings in the supporting layer structure **240**, and selectively makes the second opening (not shown) aligned to the second bit line **164** penetrate through the dielectric layer **230** and part of the cap layer **167** of the second bit line **164**, therefore, the second opening may extend into part of the cap layer **167**. In other words, the bottom surface of the second opening may be lower than the top surface of the second bit line **164**, and part of the cap layer **167** may be exposed from the second opening. Subsequently, the bottom electrode layer **251**, the capacitor dielectric layer **253** and the top electrode layer **255** are sequentially formed to form the capacitor structure **550** as shown in FIG. **13**. In particular, at least one second capacitor **550b** is aligned to the second bit line **164** in the periphery region **300b**, and extends into the part of the cap layer **167** through the bottom electrode layer **251** and directly contacts the second bit line **164**. The bottom surface of at least one second capacitor **550b** (that is, the bottom surface of the bottom electrode layer **251** filling the second opening) may be lower than the top surface of the second bit line **164**, and the at least one second capacitor **550b** only contacts the cap layer **167** to form an open circuit, thereby becoming the dummy storage node. Under this arrangement, the semiconductor memory device **500** of the present embodiment may also form a dynamic random access memory device, and the adjacent storage nodes are isolated by the second capacitor **550b** and the third capacitors **250c**, so as to maintain its overall device performance.

(33) Please refer to FIG. **14**, which shows steps of a method for forming a semiconductor memory device **600** in a fourth embodiment of the present invention. The forming steps of the front end of the semiconductor memory device **600** in the present embodiment are substantially the same as the forming steps of the front end of the semiconductor memory device **300** in the first embodiment described above, and will not be repeated here. The main difference between the present embodiment and the first embodiment is that at least one second capacitor **650b** is located above

part of the second bit line **164** and part of the contact **180** at the same time.

(34) In detail, as shown in FIG. **14**, during the process of forming the storage node pads **211**, each of the storage node pads **211** only partially overlaps with the underlying first contact **182** so that a larger process window can be obtained. Subsequently, the bottom electrode layer **251**, the capacitor dielectric layer **253** and the top electrode layer **255** are formed in sequence to form a capacitor structure **650** as shown in FIG. **14**. In this way, each of the storage node pads **211** may be located above a part of each first contact **182**, the spacer structure **170** and a part of each first bit line **162** at the same time, and the first capacitor **650a** formed later may also be located above the part of each first contact **182**, the spacer structure **170** and the part of each first bit line **162** accordingly. In addition, the second capacitor **650b** located in the periphery region **300b** may be located above a part of the second contact **184**, the spacer structure **170** and a part of the second bit line **164** at the same time. It should be noted that in the present embodiment, the etching process conditions may be further controlled to selectively make the opening (not shown) located in the peripheral region **300b** only partially penetrate the dielectric layer **230**, so as to make the bottom surface of the second capacitor **650b** (that is, the bottom surface of the bottom electrode layer **251** filling the opening) only contact the dielectric layer **230**, as shown in FIG. **14**, but not limited thereto. Under this arrangement, the semiconductor memory device **600** of the present embodiment may also form a dynamic random access memory device, and the adjacent storage nodes are isolated by the second capacitor **650b**, so as to maintain its overall device performance.

(35) In general, the present invention forms the storage node pads on the substrate through the self-aligned double patterning process or the self-aligned reverse patterning process, and forms the storage node pads only within the memory cell region to align to the contacts by controlling the overlapping parts of the patterning masks. In this way, after the capacitor structure is formed, the first capacitors that may be used as storage nodes and the second capacitor that may be used as a dummy storage node may be respectively formed, in which the storage nodes (i.e., the first capacitor) are electrically connected with the transistor device (not shown) of the semiconductor storage device through the storage node pads and the storage node plugs (i.e., the contacts) below. While there is no storage node pad under the dummy storage node (i.e., the second capacitor), instead, the dummy storage node directly contacts the cap layer of the dummy bit line, and are thereby not electrically connected with the storage node plug (i.e., the contact), so that the adjacent storage nodes may be isolated to maintain the overall device performance of the semiconductor storage device.

(36) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor memory device, comprising: a substrate, comprising a plurality of active areas and a plurality of shallow trench isolations; a plurality of bit line contacts disposed in the substrate, directly in contact with the plurality of active areas respectively; a plurality of first bit lines disposed on the plurality of bit line contacts; at least one second bit line disposed outside all of the first bit lines, wherein the plurality of first bit lines and the at least one second bit line each extends in a same extending direction, to intersect the plurality of active areas and the plurality of shallow trench isolations, each of the plurality of first bit lines comprises a first width in a direction perpendicular to the same extending direction, the at least one second bit line comprises a second width in the direction perpendicular to the same extending direction, and the second width is larger than the first width; a plurality of contacts disposed on the substrate, and alternatively and separately disposed with the first bit lines; a dielectric layer, disposed over the plurality of contacts,

the first bit lines, and the at least one second bit line; a plurality of storage node pads, disposed in the dielectric layer and respectively connected to the plurality of contacts; and a plurality of capacitors disposed on the substrate, and comprising a plurality of first capacitors and at least one second capacitor, the plurality of first capacitors being in direct contact with the plurality of storage node pads, and the at least one second capacitor being located right above the at least one second bit line.

2. The semiconductor memory device of claim 1, wherein a bottom surface of the at least one second capacitor is in direct contact with the dielectric layer.

3. The semiconductor memory device of claim 1, wherein the at least one second capacitor is located over a portion of one of the plurality of contacts and a portion of the at least one second bit line simultaneously.

4. The semiconductor memory device of claim 1, wherein each of the first bit lines and the at least one second bit line comprises a semiconductor layer, a barrier layer, a conductive layer and a cap layer sequentially stacked from bottom to top, and the at least one second capacitor is in direct contact with the cap layer of the at least one second bit line.

5. The semiconductor memory device of claim 4, wherein the at least one second capacitor is extended into the cap layer of the at least one second bit line.

6. The semiconductor memory device of claim 1, wherein a bottom surface of the at least one second capacitor is lower than a top surface of the at least one second bit line.

7. The semiconductor memory device of claim 1, further comprising at least one third capacitor respectively located corresponding to the plurality of contacts, wherein the at least one third capacitor is in direct contact with the dielectric layer.

8. The semiconductor memory device of claim 7, wherein the at least one third capacitor is disposed outside all of the first capacitors, and the at least one third capacitor is disposed between the first capacitors and the at least one second capacitor.

9. The semiconductor memory device of claim 7, wherein a bottom surface of the at least one third capacitor is lower than a top surface of the dielectric layer.

10. A semiconductor memory device, comprising: a substrate, comprising a plurality of active areas and a plurality of shallow trench isolations; a first bit line disposed on the substrate and extending in a direction, wherein the first bit line simultaneously overlaps the plurality of active areas and the plurality of shallow trench isolations in the direction; a first dielectric layer, disposed over the first bit line; a plurality of first capacitors disposed on the substrate, wherein the plurality of first capacitors simultaneously located right above the first bit line in the direction, and a bottom surface of each of the plurality of first capacitors is covered by the first dielectric layer.

11. The semiconductor memory device of claim 10, further comprising: a second dielectric layer disposed on the substrate, between the substrate and the at least one first bit line, wherein the at least one first bit line is isolated from the substrate by the second dielectric layer.

12. The semiconductor memory device of claim 11, wherein the plurality of first capacitors are separately arranged by a same pitch in the direction.

13. The semiconductor memory device of claim 11, wherein a top surface of the first dielectric layer is partially higher than a bottom surface of the plurality of first capacitors, and is partially coplanar with the bottom of the plurality of first capacitors.

14. The semiconductor memory device of claim 10, wherein bottom surfaces of the plurality of first capacitors are lower than a top surface of the first dielectric layer.

15. The semiconductor memory device of claim 10, one of the plurality of first capacitors directly contacts a top surface of the first dielectric layer, and another one of the plurality of first capacitors partially extends into the first dielectric layer.

16. The semiconductor memory device of claim 10, further comprising: a plurality of storage node pads disposed in the dielectric layer; and a plurality of second capacitors disposed on the storage node pads, in direct contact with the storage node pads respectively.

